

Ch 4 – Equations, Inequalities and Graphs

Solving modulus equations:

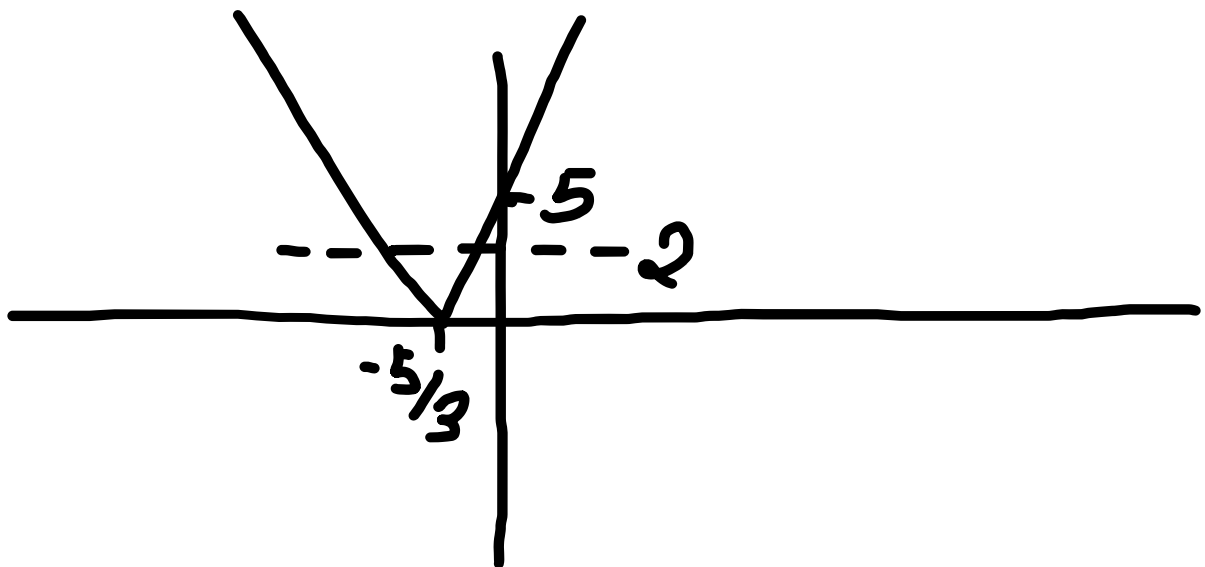
$$|3x + 5| = 2$$

$$3x + 5 = \pm 2$$

$$3x = -5 \pm 2$$

$$3x = -3, -7$$

$$x = -1, -\frac{7}{3}$$



$$|2x + 9| = 4x + 7$$

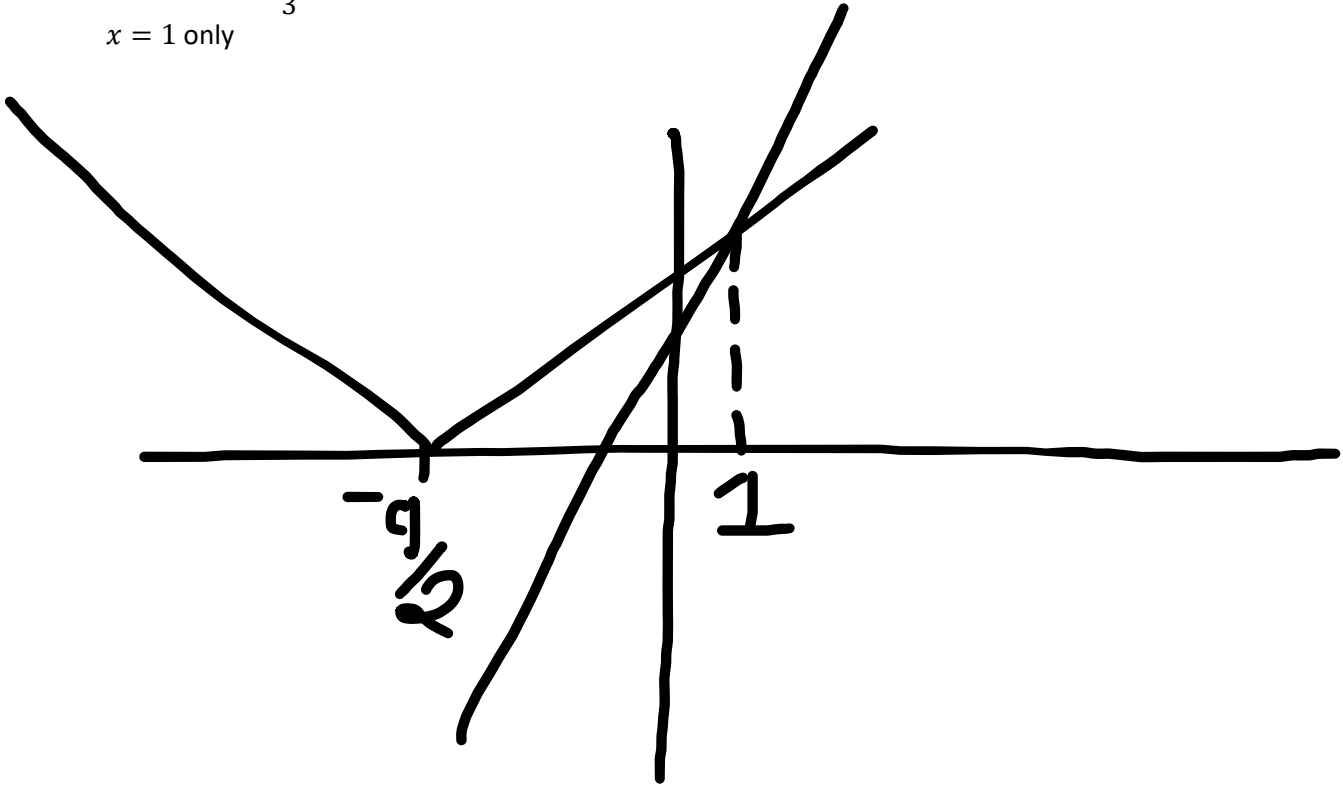
$$2x + 9 = \pm(4x + 7)$$

$$2x + 9 = 4x + 7, \quad 2x + 9 = -4x - 7$$

$$2x = 2, \quad 6x = -16$$

$$x = 1, \quad x = \frac{-8}{3}$$

$x = 1$ only



$$|x + 5| = |2x + 1|$$

$$(x + 5)^2 = (2x + 1)^2$$

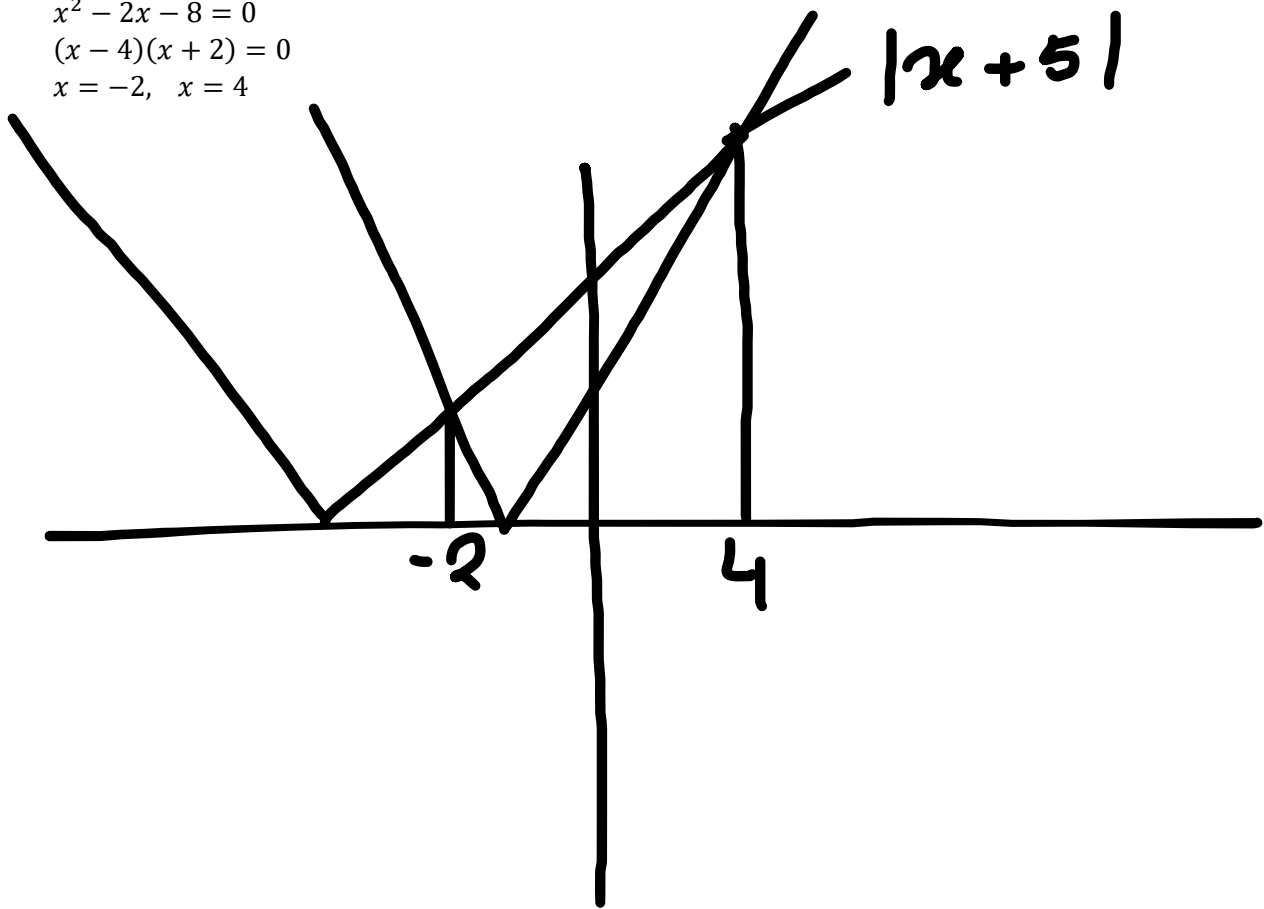
$$x^2 + 10x + 25 = 4x^2 + 4x + 1$$

$$3x^2 - 6x - 24 = 0$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$x = -2, \quad x = 4$$



$$|2x^2 + 5x + 9| = 12$$

$$2x^2 + 5x + 9 = \pm 12$$

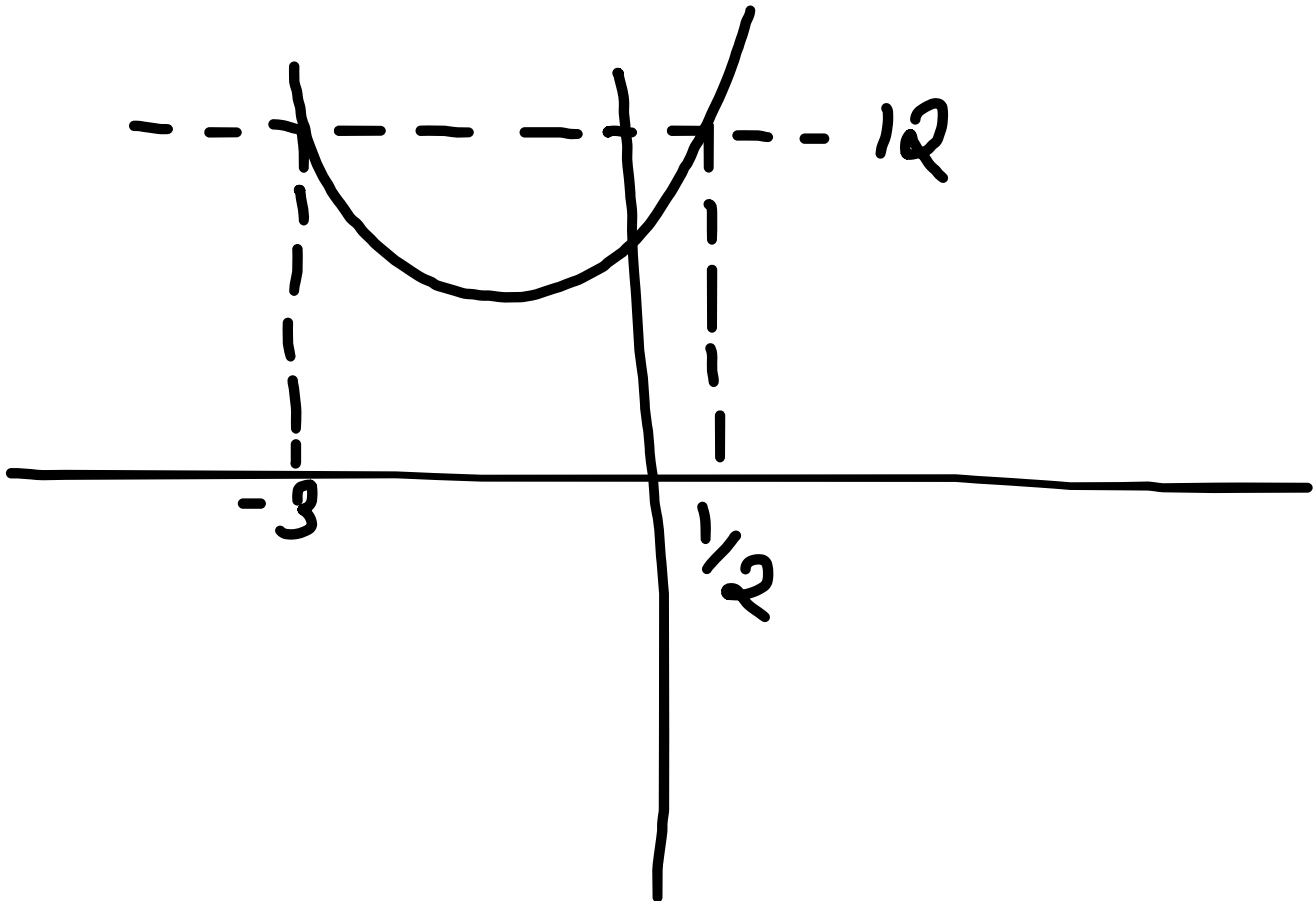
$$2x^2 + 5x - 3 = 0, \quad 2x^2 + 5x + 21 = 0$$

$$x^2 + 5x - 6 = 0, \quad x^2 + 5x + 42 = 0$$

$$(x + 6)(x - 1) = 0 \quad \text{No real solutions}$$

$$(x + 3)(2x - 1) = 0$$

$$x = -3, \quad x = \frac{1}{2}$$



Solving modulus inequalities:

$$2|x + 3| > 4$$

$$|x + 3| > 2$$

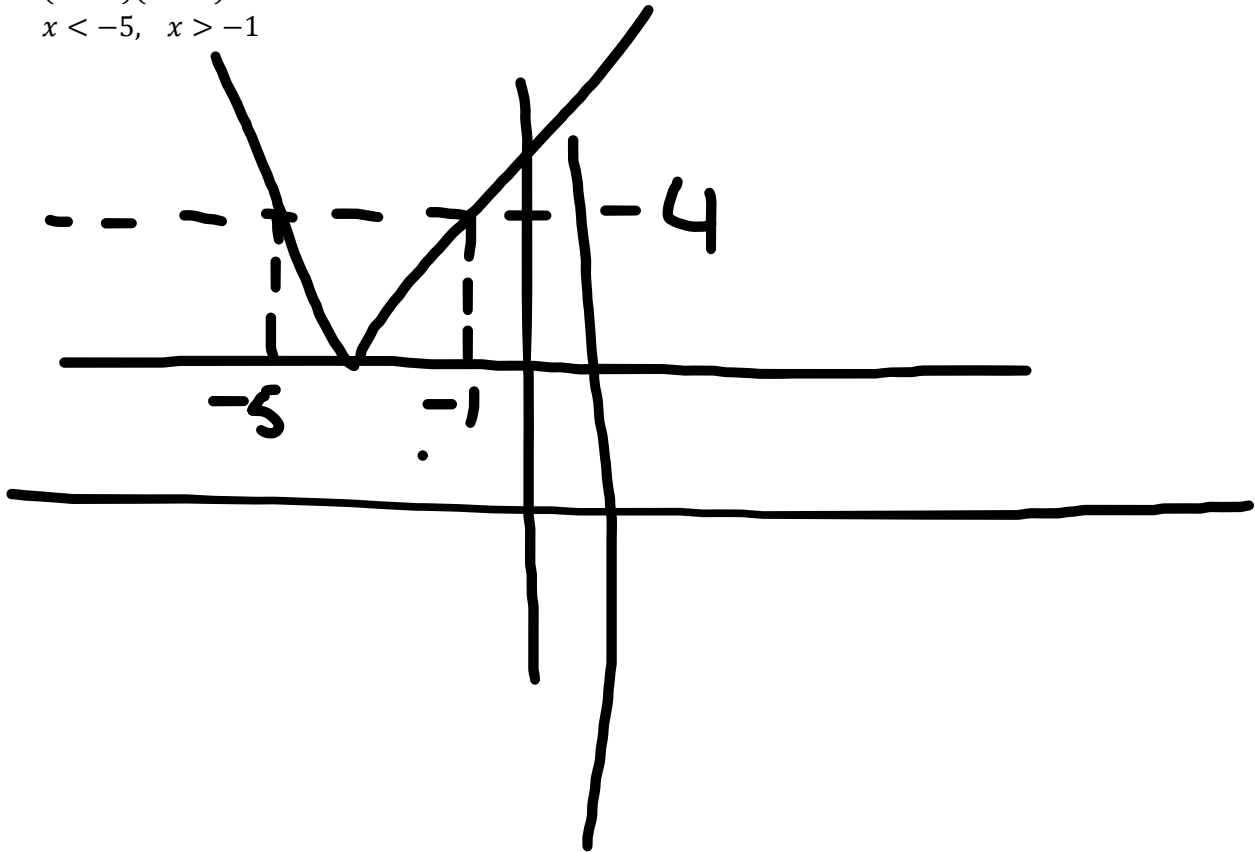
$$(x + 3)^2 > 2^2$$

$$x^2 + 6x + 9 > 4$$

$$x^2 + 6x + 5 > 0$$

$$(x + 5)(x + 1) > 0$$

$$x < -5, \quad x > -1$$



$$3|2x + 4| \leq 9$$

$$|2x + 4| \leq 3$$

$$(2x + 4)^2 \leq 3^2$$

$$4x^2 + 16x + 16 \leq 9$$

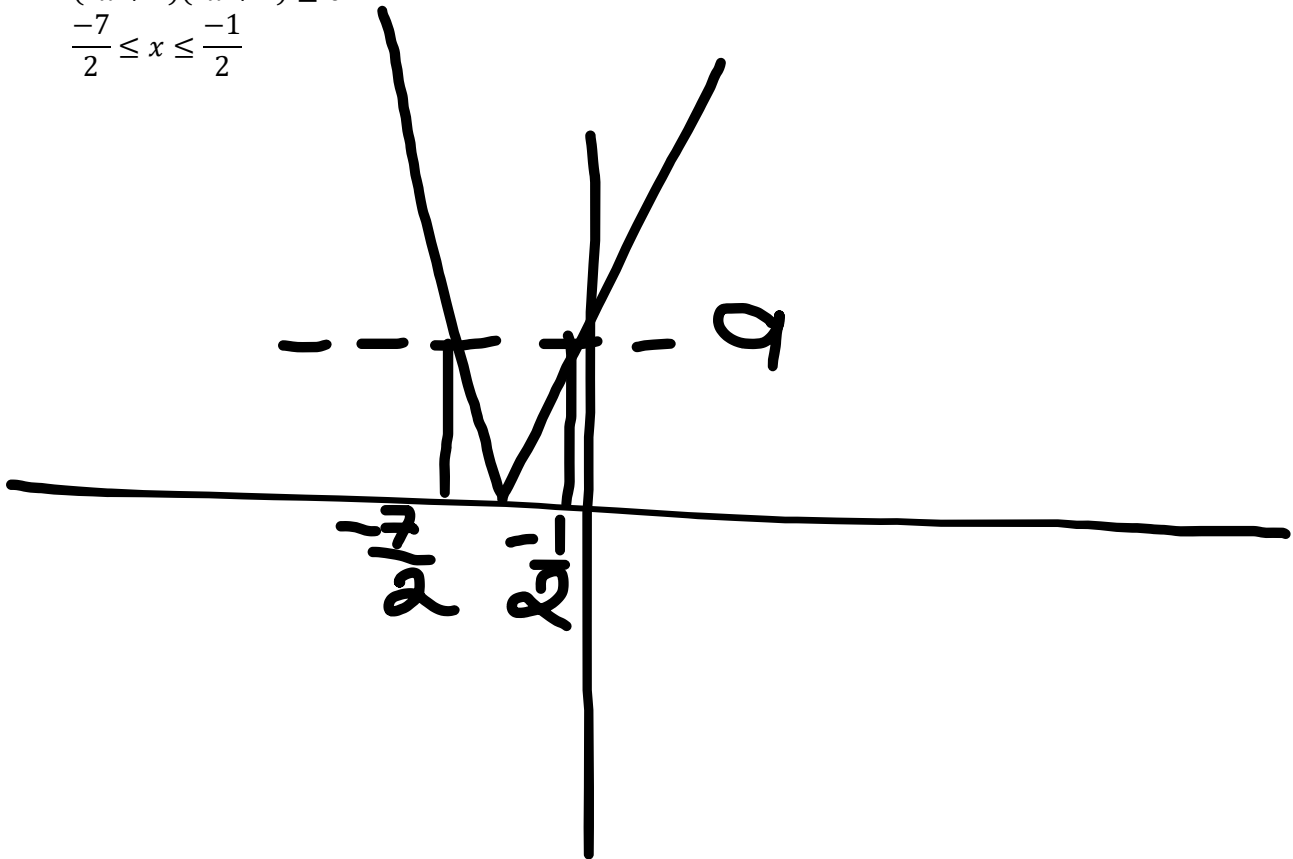
$$4x^2 + 16x + 7 \leq 0$$

$$x^2 + 4x + \frac{7}{4} \leq 0$$

$$(x + 2)(x + \frac{7}{4}) \leq 0$$

$$(2x + 7)(2x + 1) \leq 0$$

$$-\frac{7}{2} \leq x \leq -\frac{1}{2}$$



$$2|2x + 5| \leq |3x + 4|$$

$$(2(2x + 5))^2 \leq (3x + 4)^2$$

$$4(4x^2 + 20x + 25) \leq 9x^2 + 24x + 16$$

$$16x^2 + 80x + 100 \leq 9x^2 + 24x + 16$$

$$7x^2 + 56x + 84 \leq 0$$

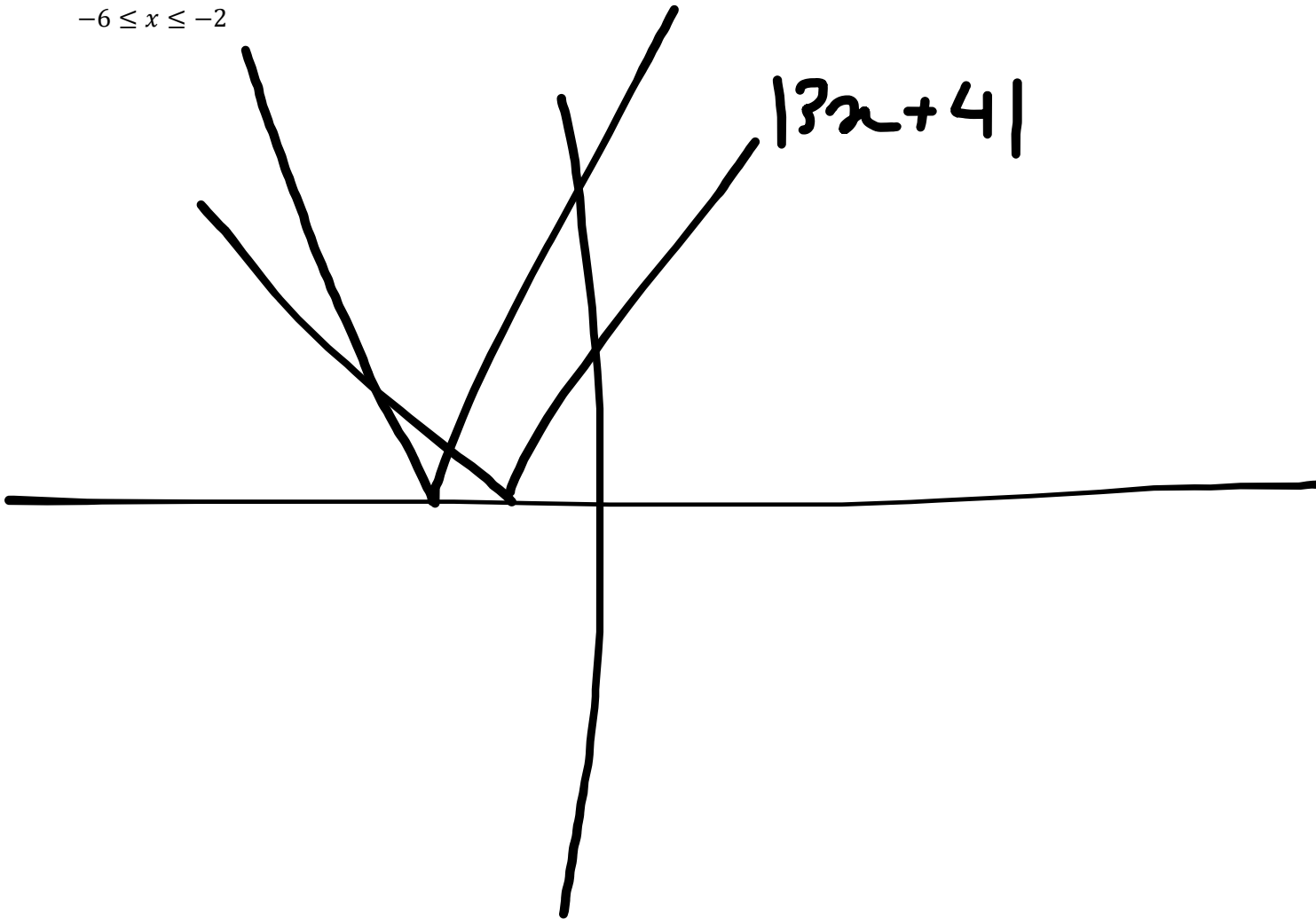
$$x^2 + 8x + 12 \leq 0$$

$$(x + 6)(x + 2) \leq 0$$

$$-6 \leq x \leq -2$$

$$2|2x + 5|$$

$$|3x + 4|$$



$$|3x + 5| \leq 2x + 5$$

$$(3x + 5)^2 \leq (2x + 5)^2$$

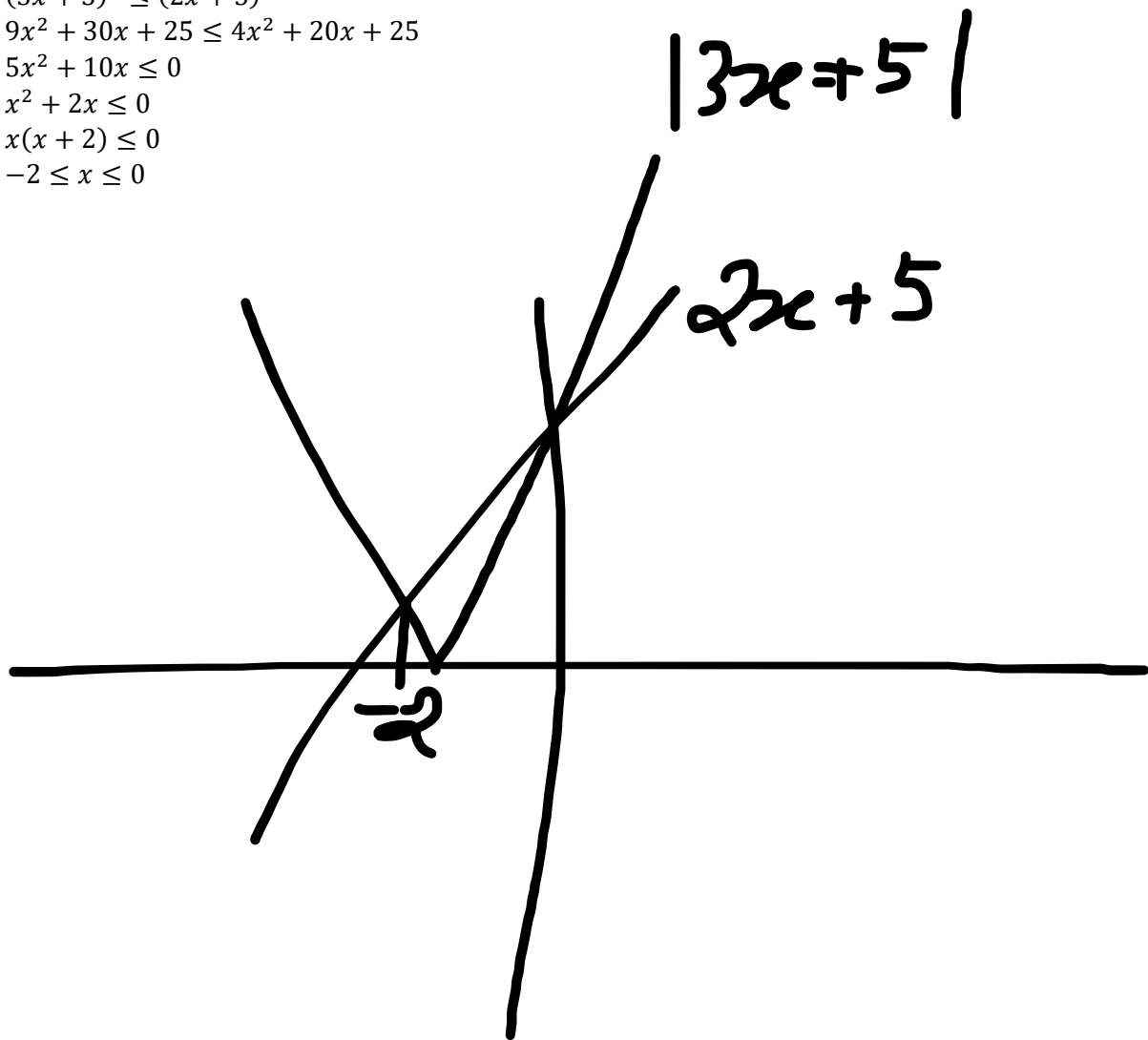
$$9x^2 + 30x + 25 \leq 4x^2 + 20x + 25$$

$$5x^2 + 10x \leq 0$$

$$x^2 + 2x \leq 0$$

$$x(x + 2) \leq 0$$

$$-2 \leq x \leq 0$$



$$|2x^2 + 7x + 3| > 3$$

$$2x^2 + 7x + 3 > 3$$

$$2x^2 + 7x > 0$$

$$x(2x + 7) > 0$$

$$x < \frac{-7}{2}, x > 0$$

$$2x^2 + 7x + 3 < -3$$

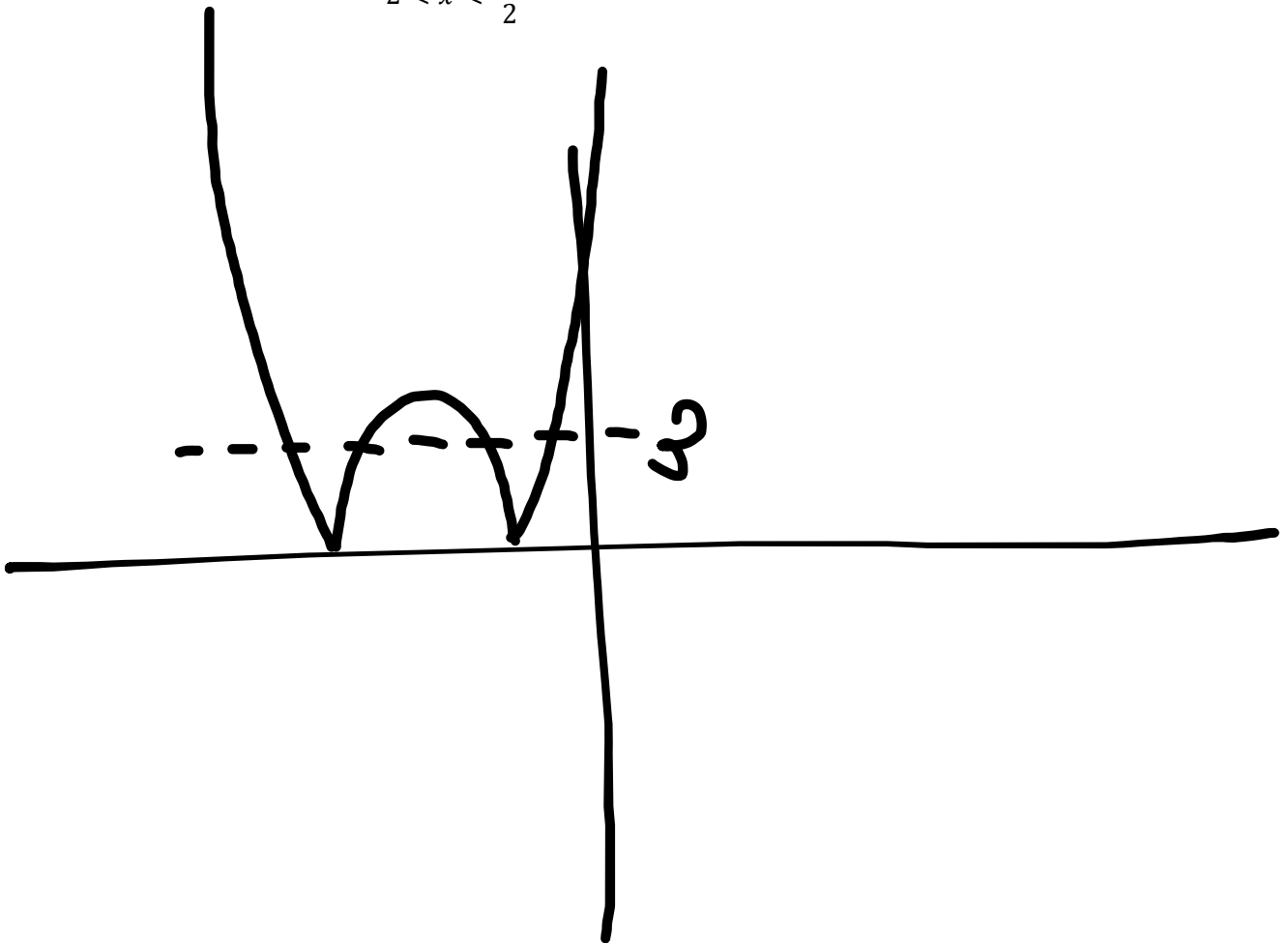
$$2x^2 + 7x + 6 < 0$$

$$x^2 + 7x + 12 < 0$$

$$(x + 3)(x + 4) < 0$$

$$(2x + 3)(x + 2) < 0$$

$$-2 < x < \frac{-3}{2}$$



$$|x^2 + 3x + 4| \leq 2$$

$$x^2 + 3x + 4 \leq 2$$

$$x^2 + 3x + 2 \leq 0$$

$$(x + 2)(x + 1) \leq 0$$

$$-2 \leq x \leq 1$$

$$x^2 + 3x + 4 \geq -2$$

$$x^2 + 3x + 6 \geq 0$$

No real solutions

Substituting to form Quadratic Equations:

Eg: $x^{\frac{4}{3}} + x^{\frac{2}{3}} - 12 = 0$

$$a = x^{\frac{2}{3}}$$

$$a^2 + a - 12 = 0$$

$$(a + 4)(a - 3) = 0$$

$$a = -4, a = 3$$

$$x^{\frac{2}{3}} = -4, 3$$

$$x^{\frac{2}{3}} = 3$$

$$x = \pm 3\sqrt{3}$$

Eg: $2(\ln 5x)^2 + \ln 5x - 6 = 0$

$$a = \ln 5x$$

$$2a^2 + a - 6 = 0$$

$$(a + 2)(2a - 3) = 0$$

$$a = -2, \frac{3}{2}$$

$$\ln 5x = -2, \frac{3}{2}$$

$$5x = e^{-2}, e^{\frac{3}{2}}$$

$$x = \frac{e^{-2}}{5}, \quad x = \frac{e^{\frac{3}{2}}}{5}$$

Eg: $3e^x = 12 - 5e^{-x}$

$$3e^x + \frac{5}{e^x} - 12 = 0$$

$$3(e^x)^2 - 12e^x + 5 = 0$$

$$a = e^x$$

$$3a^2 - 12a + 5 = 0$$

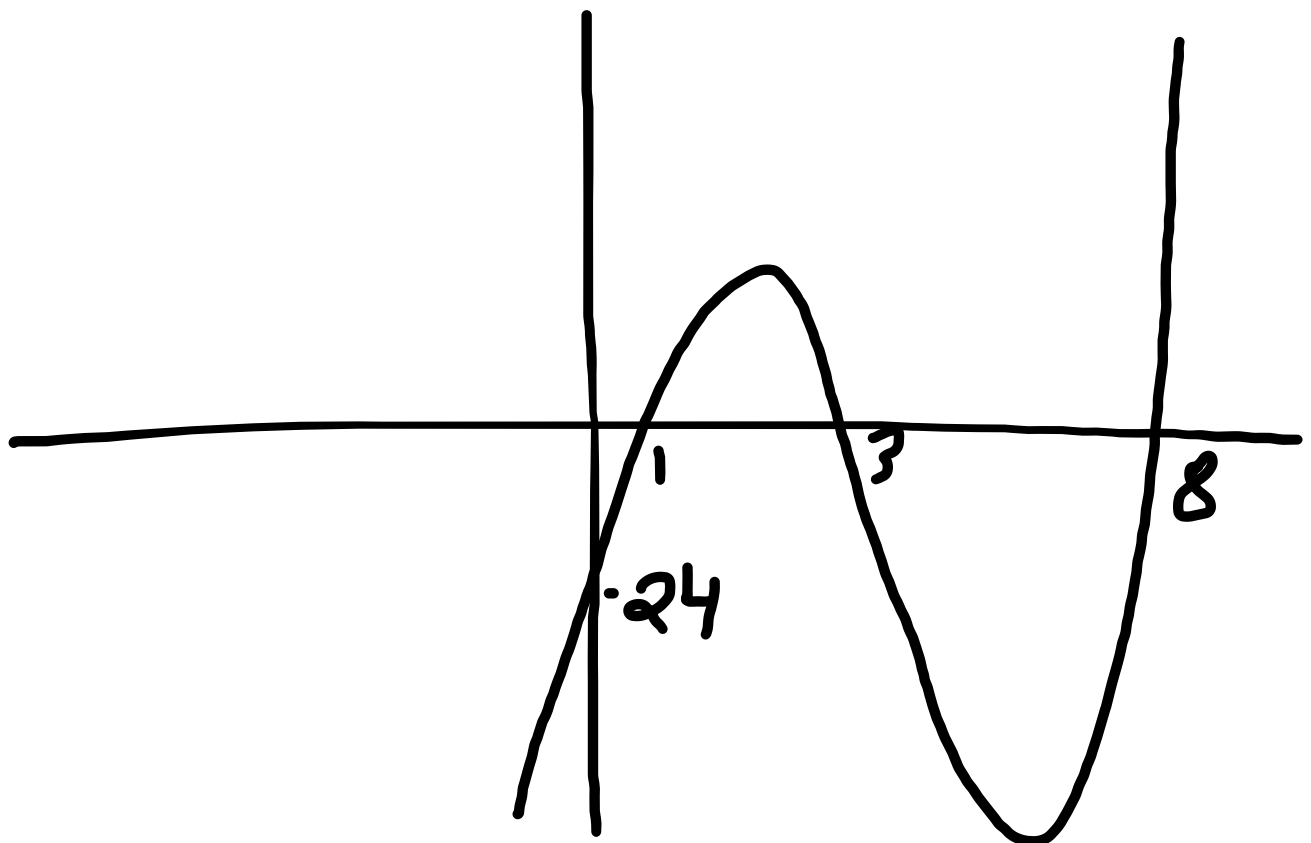
$$a = \frac{6 \pm \sqrt{21}}{3}$$

$$e^x = \frac{6 \pm \sqrt{21}}{3}$$

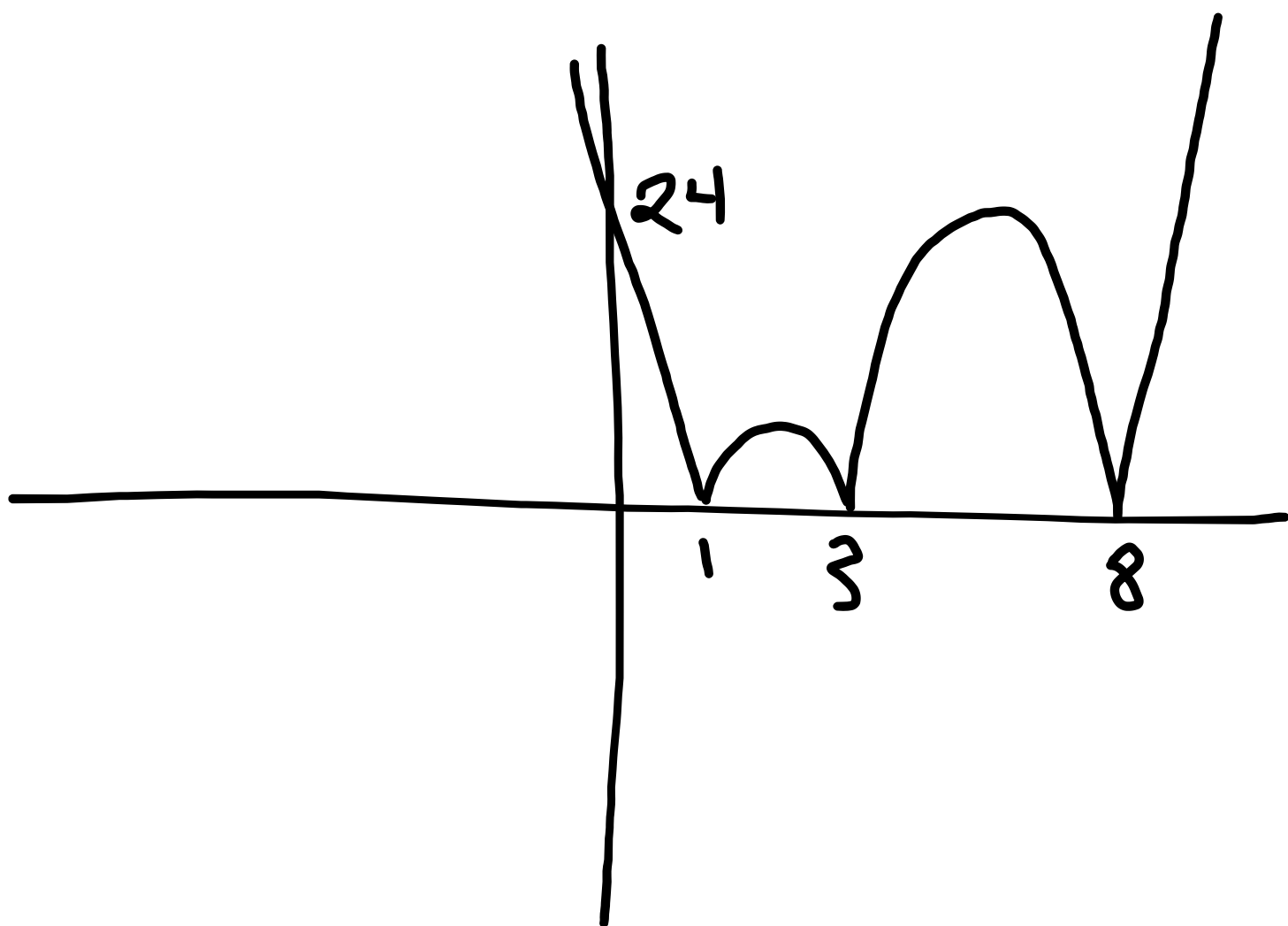
$$x = \ln \frac{6 \pm \sqrt{21}}{3}$$

Graphs of cubic polynomials and their moduli:

Eg: $f(x) = (x - 1)(x - 3)(x - 8)$

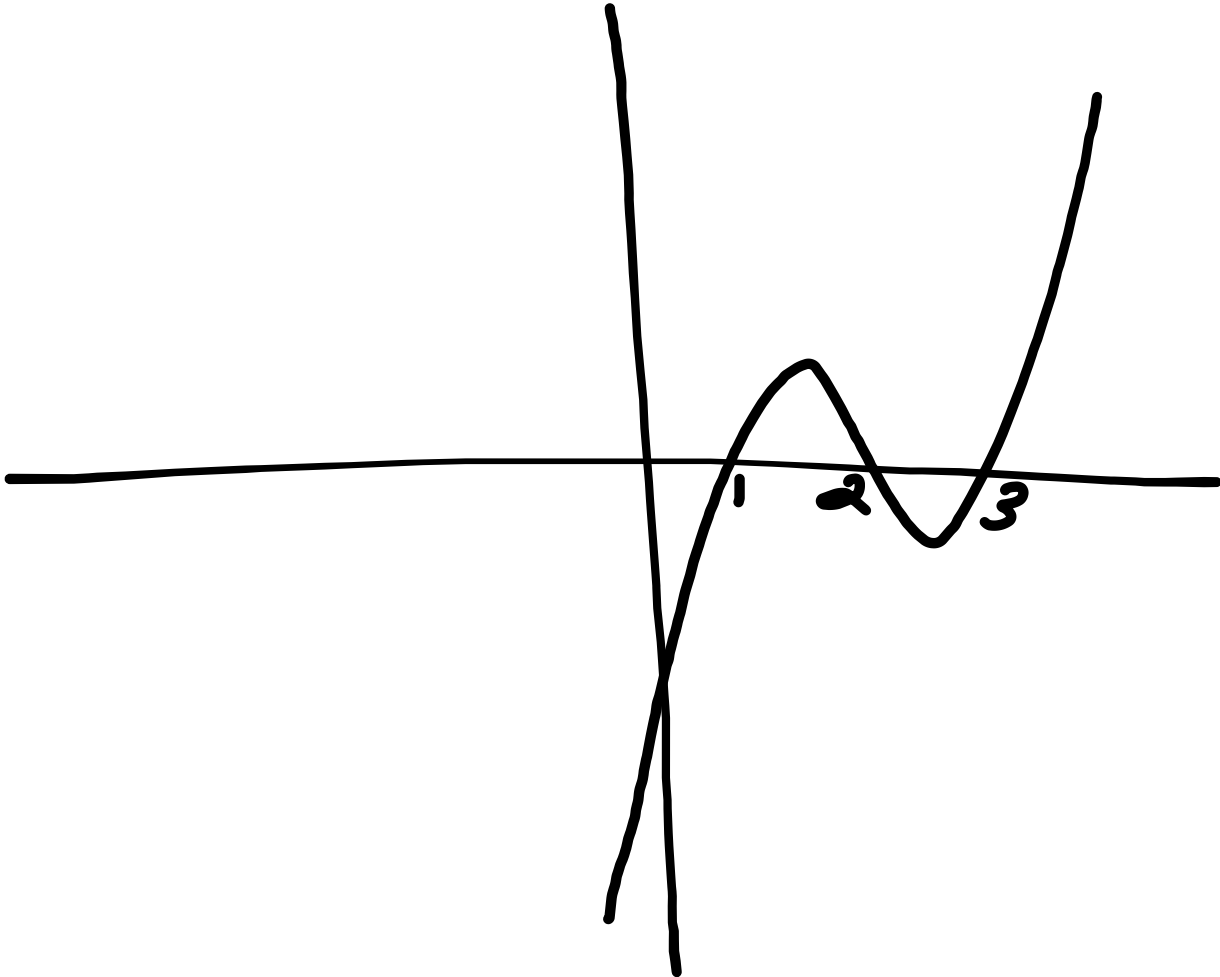


Eg: $|f(x) = (x - 1)(x - 3)(x - 8)|$



Cubic inequalities:

Eg: $f(x) = (x - 1)(x - 2)(x - 3), f(x) > 0$



$$1 < x < 2$$

$$x > 3$$